17.3: Eukaryotic cells modify RNA after transcription

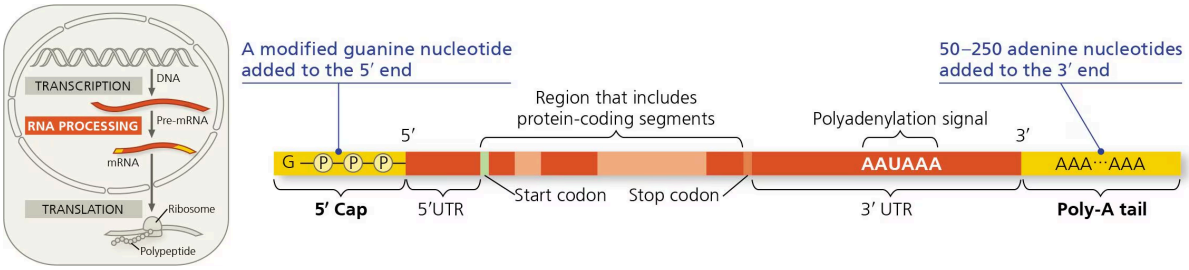
Enzymes in the eukaryotic nucleus modify pre-mRNA in specific ways before the genetic message is dispatched to the cytoplasm. During this **RNA processing**, both ends of the primary transcript are altered. Also, in most cases, certain interior sections of the RNA molecule are cut out and the remaining parts spliced together. These modifications produce an mRNA molecule ready for translation.

Alteration of mRNA Ends

Each end of a pre-mRNA molecule is modified in a particular way **(Figure 17.11)**. The 5′ end, which is synthesized first, receives a **5′ cap**, a modified form of a guanine (G) nucleotide added onto the 5′ end after transcription of the first 20–40 nucleotides have been transcribed. The 3′ end of the pre-mRNA molecule is also modified before the mRNA exits the nucleus. Recall that the pre-mRNA is cut and released soon after the polyadenylation signal, AAUAAA, is transcribed. At the 3′ end, an enzyme then adds 50–250 more adenine (A) nucleotides, forming a **poly-A tail**. The 5′ cap and poly-A tail share several important functions. First, they seem to facilitate the export of the mature mRNA from the nucleus. Second, they help protect the mRNA from degradation by hydrolytic enzymes. And third, they help ribosomes attach to the 5′ end of the mRNA once it reaches the cytoplasm. In addition to the cap and tail on a eukaryotic mRNA molecule, Figure 17.11 shows the untranslated regions (UTRs) at the 5′ and 3′ ends of the mRNA (referred to as the 5′ UTR and 3′ UTR). The UTRs are parts of the mRNA that will not be translated into protein, but they have other functions, such as ribosome binding.

Figure 17.11

RNA processing: addition of the 5′ cap and poly-A tail.

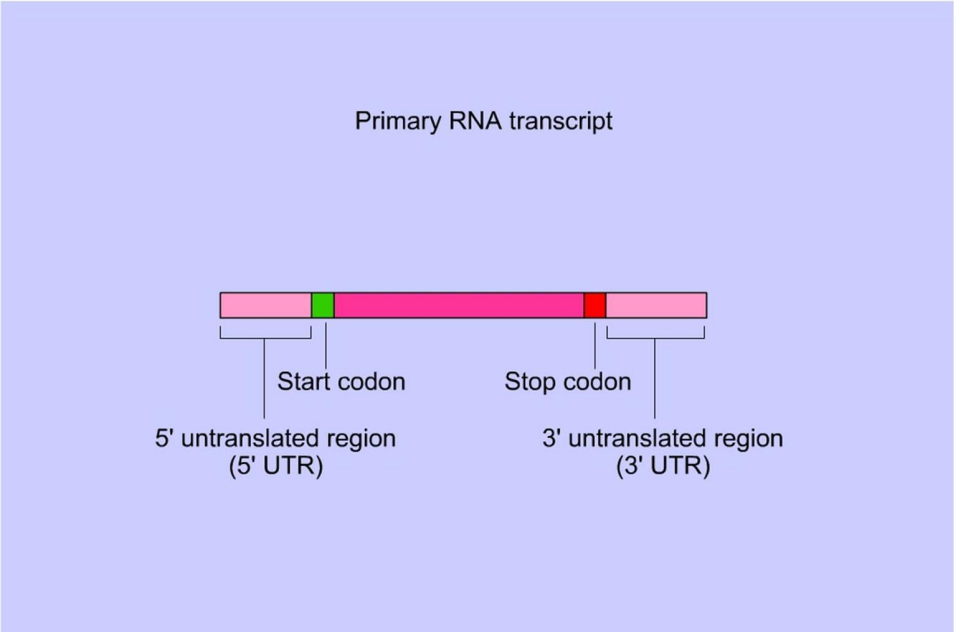


Overview: In the nucleus, transcription of D N A results in pre-m R N A. R N A processing leads to m R N A, which leaves the nucleus. Translation of the m R N A takes place in the cytoplasm on ribosomes, where polypeptides are synthesized. R N A molecule diagramed from left to right (5 prime to 3 prime): 5 prime cap (G plus 3 phosphate groups); 5 prime UTR; region with protein-coding segments (between a start and stop codon); 3 prime-U T R with polyadenylation signal with sequence A A U A A A; Poly-A tail with 50-250 adenine nucleotides added to 3 prime end.

Enzymes modify the two ends of a eukaryotic pre-mRNA molecule. The modified ends may promote the export of mRNA from the nucleus, and they help protect the mRNA from degradation. When the mRNA reaches the cytoplasm, the modified ends, in conjunction with certain cytoplasmic proteins, help with ribosome attachment. The 5′ cap and poly-A tail are not translated into protein, nor are the regions called the 5′ untranslated region (5′UTR) and 3′ untranslated region (3′UTR). The pink segments are introns, which will be described shortly (see Figure 17.12).

Animation

Adding the 5′ Cap and Poly-A Tail



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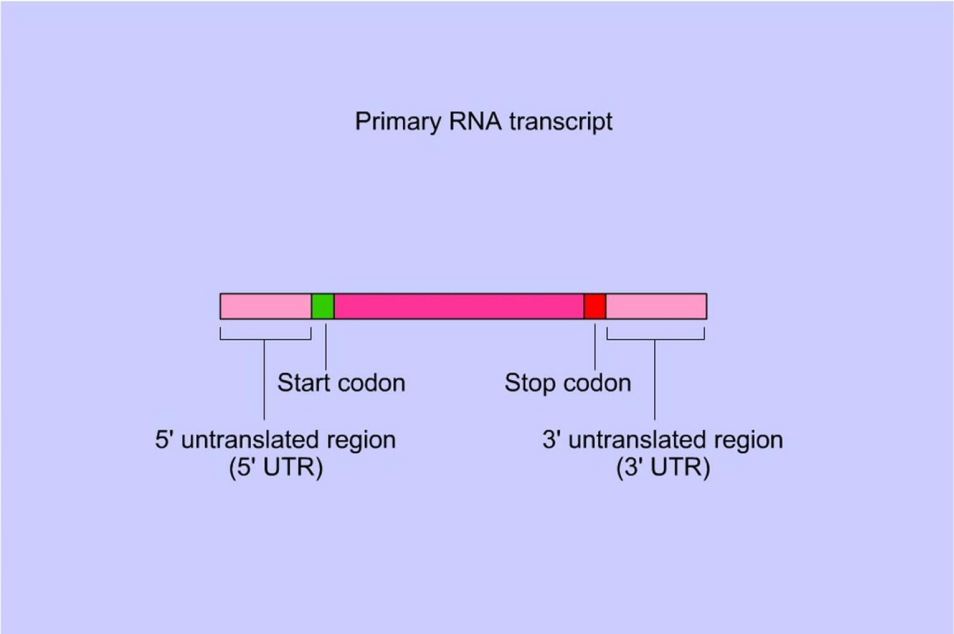

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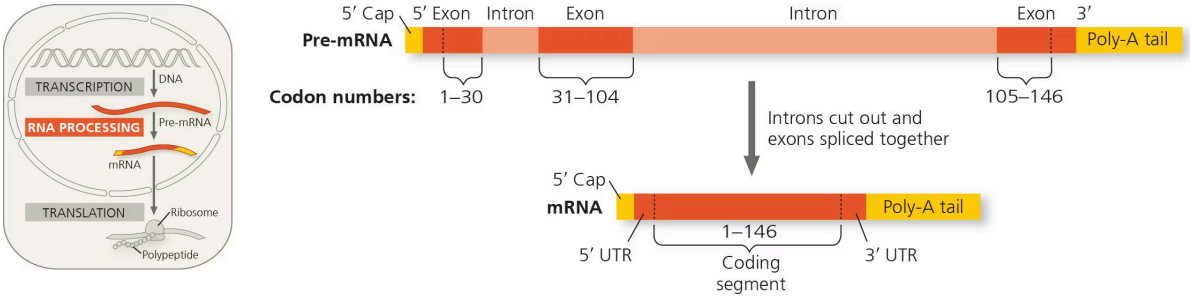
Split Genes and RNA Splicing

A remarkable stage of RNA processing in the eukaryotic nucleus is called **RNA splicing(Figure 17.12)**, in which large portions of the RNA primary transcript molecules are removed and the remaining portions are reconnected. This cut-and-paste job is similar to editing a movie. The average length of a transcription unit along a human DNA molecule is about 27,000 nucleotide pairs, so the primary RNA transcript is also that long. However, the average-sized protein of 400 amino acids requires only 1,200 nucleotides in RNA to code for it. (Remember, each amino acid is encoded by a *triplet* of nucleotides.) This is because most eukaryotic genes and their RNA transcripts have long noncoding stretches of nucleotides, regions that are not translated. Even more surprising is that most of these noncoding sequences are interspersed between coding segments of the gene and thus between coding segments of the pre-mRNA. In other words, the sequence of DNA nucleotides that codes for a eukaryotic polypeptide is usually not continuous; it is split into segments. The noncoding segments of nucleic acid that lie between coding regions are called intervening sequences, or **introns**. The other regions are called **exons**, because they are eventually expressed, usually by being translated into amino acid sequences. (Exceptions include the UTRs of the exons at the ends of the RNA, which make up part of the mRNA but are not translated into protein. Because of these exceptions, you may prefer to think of exons as sequences of RNA that *ex*it the nucleus.) The terms *intron* and *exon* are used for both RNA sequences and the DNA sequences that specify them.

Figure 17.12

RNA processing: RNA splicing.



Overview: In the nucleus, transcription of D N A results in pre-m R N A. R N A processing leads to m R N A, which leaves the nucleus. Translation of the m R N A takes place in the cytoplasm on ribosomes, where polypeptides are synthesized. Pre-m R N A labeled as follows: 5 prime cap; 5 prime U T R; Exon (codon numbers 1-30); short intron; exon (codon numbers 31-104); long intron; exon (codon numbers 105-146); 3 prime U T R; poly-A tail. Arrows flow to result after introns cut out and exons spliced together: m R N A: 5 prime cap; 5 prime U T R; Coding segment 1-146; 3 prime U T R; poly-A tail.

The RNA molecule shown here codes for β-globin, one of the polypeptides of hemoglobin. The numbers under the RNA refer to codons; β-globin is 146 amino acids long. The β-globin gene and its pre-mRNA transcript have three exons, corresponding to sequences that will exit the nucleus as mRNA. (The 5′ UTR and 3′ UTR do not code for proteins, but they are parts of exons because they remain in the mRNA.) During RNA processing, the introns are cut out and the exons spliced together. In many genes, the introns are much longer than the exons.

DRAW IT *On the mRNA, indicate the sites of the start and stop codons.*

*For suggested answer, see*Appendix A*.*

BioFlix® Animation

RNA Processing

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In making a primary transcript from a gene, RNA polymerase II transcribes both introns and exons from the DNA, but the mRNA molecule that enters the cytoplasm is an abridged version. In the process of RNA splicing, the introns are cut out from the molecule and the exons joined together, forming an mRNA molecule with a continuous coding sequence.

How is pre-mRNA splicing carried out? The removal of introns is accomplished by a large complex made of proteins and small RNAs called a **spliceosome**. This complex binds to several short nucleotide sequences along an intron, including key sequences at each end **(Figure 17.13)**. The intron is then released (and rapidly degraded), and the spliceosome joins together the two exons that flanked the intron. The small RNAs in the spliceosome not only participate in spliceosome assembly and splice site recognition, but also catalyze the splicing reaction; like proteins, RNAs can act as catalysts.

Figure 17.13

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A spliceosome splicing a pre-mRNA.
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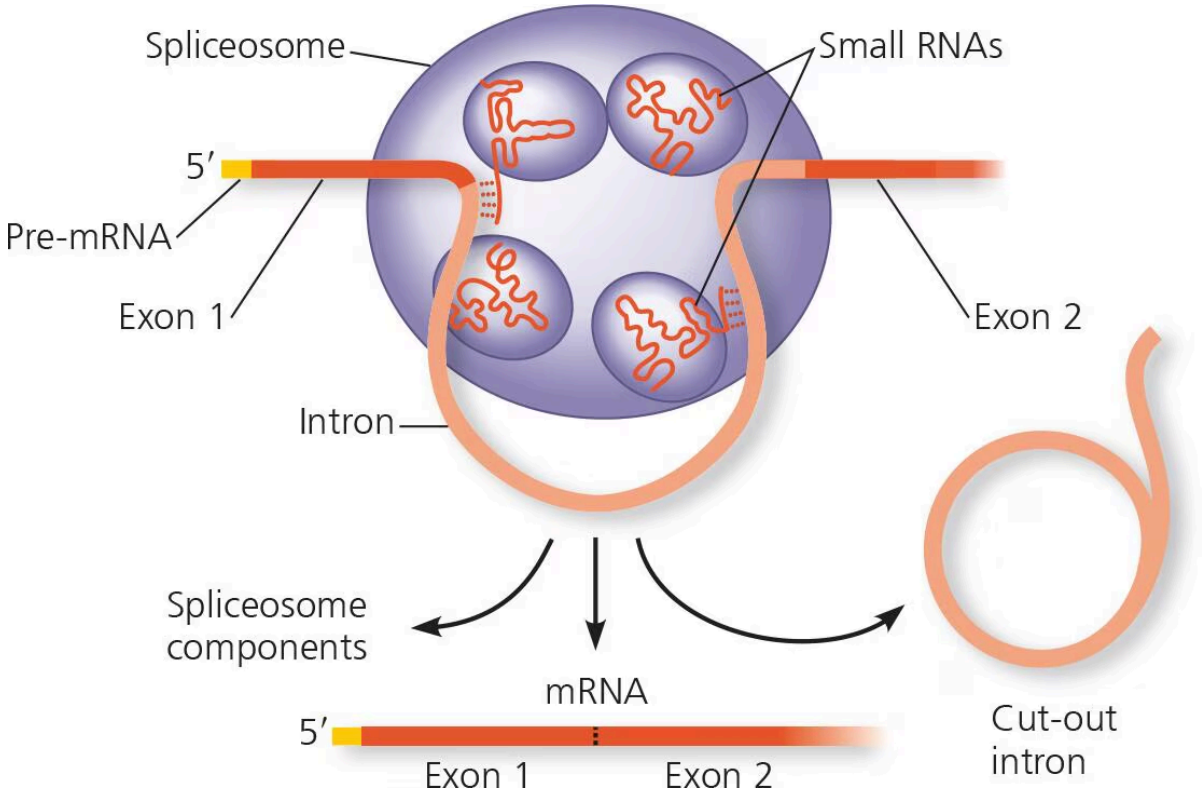
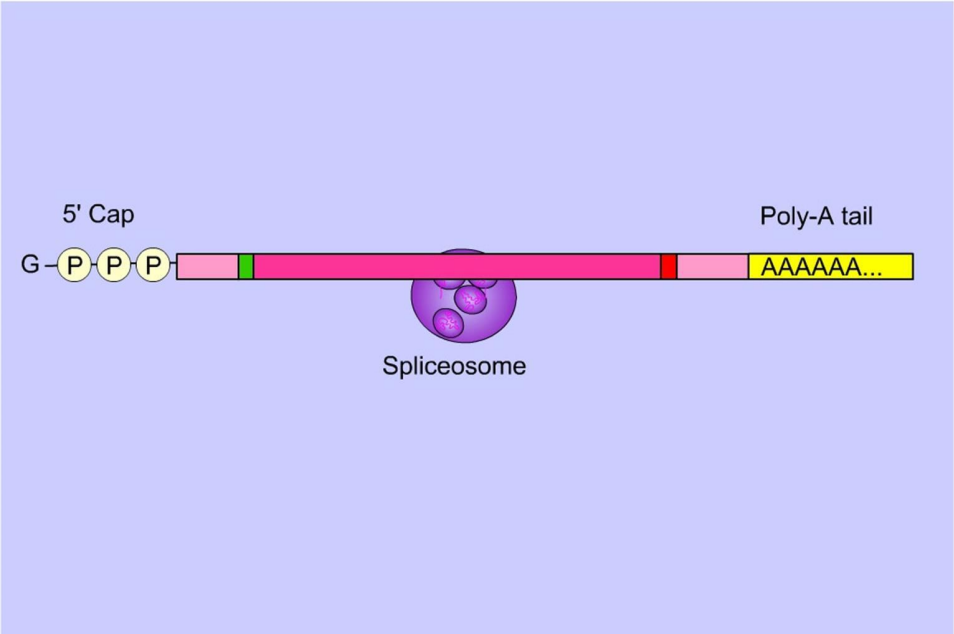


The diagram shows a portion of a pre-mRNA transcript, with an intron (pink) flanked by two exons (red). Small RNAs within the spliceosome base-pair with nucleotides at specific sites along the intron. Next, small spliceosome RNAs catalyze cutting of the pre-mRNA and the splicing together of the exons, releasing the intron for rapid degradation.

Pre-m R N A drawn with 5 prime end, Exon 1, long intron and Exon 2 labeled. Spliceosome shown over long intron section, which forms base pairs (dashed lines) with some of the many small R N As inside the spliceosome. Arrows indicate that the intron is spliced out and released, while the exons are spliced together. m R N A molecule results with regions labeled 5 prime, Exon 1, Exon 2.

Animation

A Spliceosome



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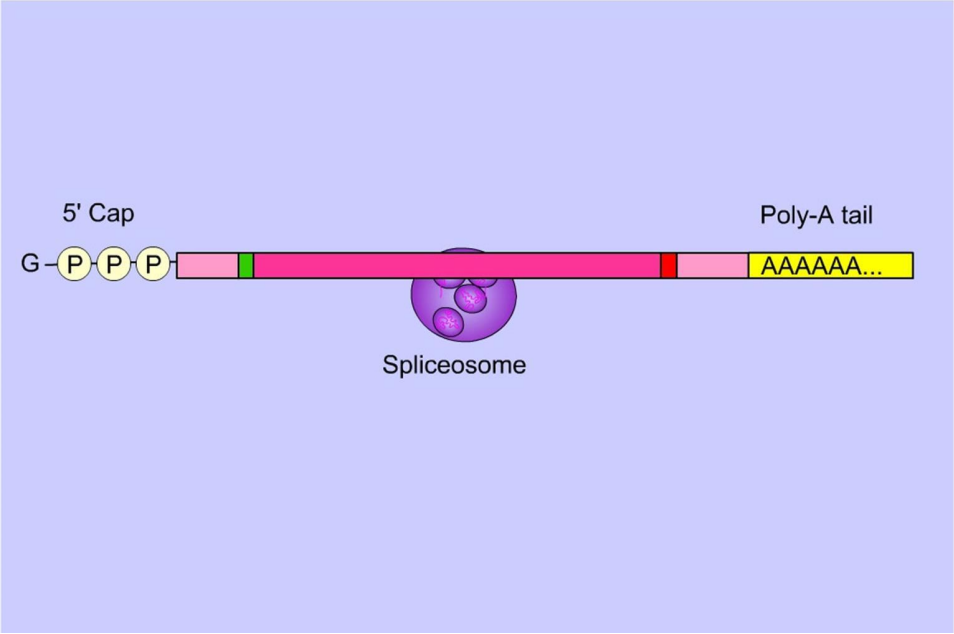

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Interview

Interview with Joan Steitz: Studying RNA, her "favorite molecule"

Ribozymes

The idea of a catalytic role for the RNAs in the spliceosome arose from the discovery of **ribozymes**, RNA molecules that function as enzymes. In some organisms, RNA splicing can occur without proteins or even additional RNA molecules: The intron RNA functions as a ribozyme and catalyzes its own removal. For example, in the ciliate protist *Tetrahymena*, self-splicing occurs in the production of ribosomal RNA (rRNA), a component of the organism's ribosomes. The pre-rRNA actually removes its own introns! The discovery of ribozymes invalidated the idea that all biological catalysts are proteins.

Three properties of RNA enable some RNA molecules to function as enzymes. First, because RNA is single-stranded, a region of an RNA molecule may base-pair, in an antiparallel arrangement, with a complementary region elsewhere in the same molecule; this gives the molecule a particular three-dimensional structure. A specific structure is essential to the catalytic function of ribozymes, just as it is for enzymatic proteins. Second, like certain amino acids in an enzymatic protein, some of the bases in RNA contain functional groups that can participate in catalysis. Third, the ability of RNA to hydrogen-bond with other nucleic acid molecules (either RNA or DNA) adds specificity to its catalytic activity. For example, complementary base pairing between the RNA of the spliceosome and the RNA of a primary RNA transcript precisely locates the region where the ribozyme catalyzes splicing. Later in this chapter, you will see how these properties of RNA also allow it to perform important noncatalytic roles in the cell, such as recognition of the three-nucleotide codons on mRNA.

The Functional and Evolutionary Importance of Introns

EVOLUTION Whether or not RNA splicing and the presence of introns have provided selective advantages during evolutionary history is a matter of some debate. In any case, it is informative to consider their possible adaptive benefits. Specific functions have not been identified for most introns, but at least some contain sequences that regulate gene expression, and many affect gene products.

One important consequence of the presence of introns in genes is that a single gene can encode more than one kind of polypeptide. Many genes are known to give rise to two or more different polypeptides, depending on which segments are treated as exons during RNA processing; this is called **alternative RNA splicing** (see Figure 18.14). Results from the Human Genome Project (discussed in Concept 21.1) suggest that alternative RNA splicing is one reason humans can get along with about the same number of genes as a nematode (roundworm). Because of alternative splicing, the number of different protein products an organism produces can be much greater than its number of genes.

Proteins often have a modular architecture consisting of discrete structural and functional regions called **domains**. One domain of an enzyme, for example, might include the active site, while another might allow the enzyme to bind to a cellular membrane. In quite a few cases, different exons code for the different domains of a protein **(Figure 17.14)**.

Figure 17.14

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Correspondence between exons and protein domains.
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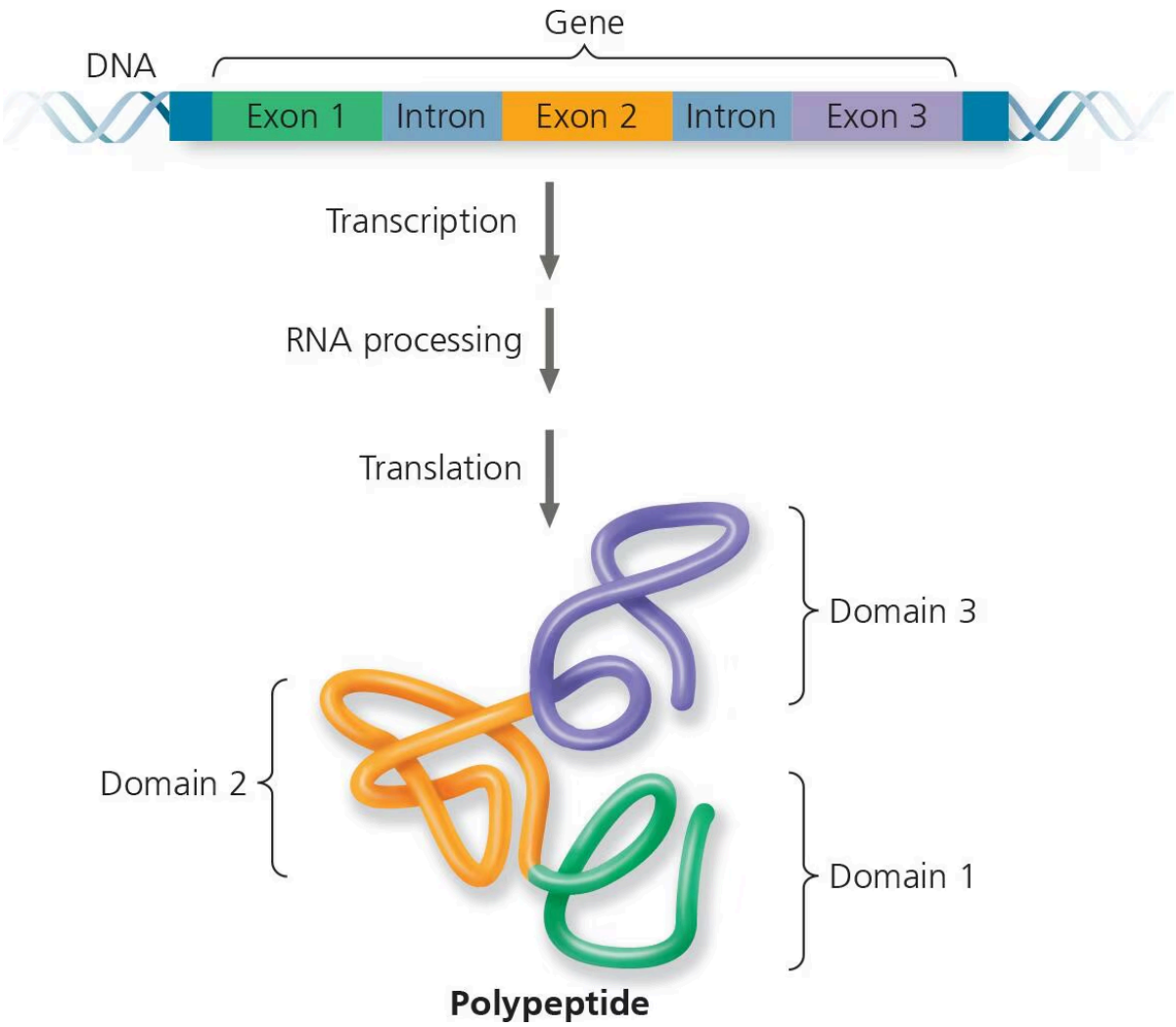

Gene labeled on D N A; regions of gene are: Exon 1; Intron; Exon 2; Intron; Exon 3. Following Transcription, R N A Processing and Translation, in the polypeptide Domain 1 corresponds to Exon 1; Domain 2 corresponds to Exon 2 and Domain 3 corresponds to Exon 3.

The presence of introns in a gene may facilitate the evolution of new and potentially beneficial proteins as a result of a process known as *exon shuffling* (see Figure 21.16). Introns increase the probability of crossing over between the exons of alleles of a gene—simply by providing more terrain for crossovers without interrupting coding sequences. This might result in new combinations of exons and proteins with altered structure and function. We can also imagine the occasional mixing and matching of exons between completely different (nonallelic) genes. Exon shuffling of either sort could lead to new proteins with novel combinations of functions. While most of the shuffling would result in nonbeneficial changes, occasionally a beneficial variant might arise.